



MST400 – Project 2

Secure P2S VPN Connection
Cloud Network

Summer 2026

What is this Project About?

In this project, you will learn how to provide **secure remote access** to Azure resources using a **Point-to-Site (P2S) VPN**. You will configure an Azure VPN Gateway, install a VPN client on your local computer, and demonstrate secure SSH connectivity to a Linux VM in Azure using **only the VM's private IP address**.

You will also create a **PowerPoint presentation** and record a short demo video explaining your solution.

What You Will Learn

By completing this project, you will learn how to:

- Create Azure infrastructure (Resource Groups, VNets, Subnets, VMs)
- Deploy and configure an **Azure VPN Gateway**
- Enable **Point-to-Site VPN** with a client address pool
- Download and install a **VPN client** on your local PC
- Connect securely to a Linux VM using **SSH over a private IP**
- Create a PowerPoint presentation and demonstrate your deployed solution

Project Scenario

You are working as a Cloud Administrator.

Your company has a Linux server hosted in Azure. Remote employees need **secure access** to this server from their home computers. Public access to the server is not allowed for security reasons.

Your task is to:

- Deploy a **Point-to-Site VPN Gateway** in Azure (Canada Central region)
- Configure your **local computer** to connect to Azure through the VPN
- Demonstrate that you can **SSH to the Linux VM** using its **private IP** only when the VPN is connected

Architecture Diagram



What Your Final Solution Must Do

✓ VPN Connectivity

- Your local computer must connect to Azure via **Point-to-Site VPN**
- VPN client must show status: **Connected**

✓ Secure SSH Access

- You must SSH to the Linux VM using its **private IP address**
- SSH must work **only when the VPN is connected**
- When VPN is disconnected, SSH to the private IP must **fail**

✓ Proper Configuration

- VPN Gateway must be properly configured with:
 - Client address pool (non-overlapping with VNet)
 - Tunnel type (OpenVPN/IKEv2)
 - Authentication method (Certificate-based authentication)
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Technical Requirements

Resource Naming (VERY IMPORTANT)

Use your student ID or name as a prefix for all resources.

Example:

- Resource Group → *YourName-MST400S26-R1*
- VNet → *YourName-VNet-CanadaCentral*
- VPN Gateway → *YourName-VpnGw*
- Linux VM → *YourName-P2S-VM*

Virtual Network and Subnets

Region: Canada Central

VNet Configuration:

- Use **ONLY your assigned personal IP address space**
- Delete the default subnet and create new subnets:
 - GatewaySubnet (required for VPN Gateway)
 - WorkloadSubnet (for the Linux VM)

VPN Gateway Configuration

- Gateway type: **VPN**
- VPN type: **Route-based**
- SKU: As specified in class (e.g., VpnGw1 or Basic)
- **Point-to-Site configuration:**
 - Client address pool: (e.g., 172.16.0.0/24 – must NOT overlap with VNet)
 - Tunnel type: OpenVPN or IKEv2
 - Authentication: Certificate-based

Linux VM

- Name: *YourName-P2S-VM*
- Deploy in WorkloadSubnet
- Authentication type: Generate your key pair
- SSH public key source: (Use existing public key)
- Enable SSH (port 22)
- NSG must allow SSH from the **P2S client address pool**
- Note the **private IP address** of this VM

Local PC Requirements

- Install VPN client / import VPN profile from Azure
- Connect to the VPN
- Use SSH client (Terminal/PowerShell/PuTTY) to connect to the VM

Networking Rules

- Use **ONLY your assigned personal IP address space**
- Delete the default subnet in the VNet and create new subnets using your assigned IP address space
- Use **ONLY your assigned Azure odl_user account for this course**

PowerPoint Presentation Requirement (VERY IMPORTANT)

You must create a PowerPoint presentation that includes:

- Your architecture design diagram showing IPs and resources
- Screenshots of all provisioned Azure resources
- Screenshots of:
 - VNet and subnets with IP address spaces
 - VPN Gateway configuration
 - Point-to-Site configuration (client address pool, tunnel type)
 - Certificates generated and used
 - Linux VM configuration

- Explanation of how Point-to-Site VPN works
- Explanation of your testing approach

You will use this PowerPoint during your video presentation and submit it with your video link.

Project Grading

Part 1 — Resource Setup (25 Points)

You must create and show:

- Resource Group
 - Virtual Network
 - Subnetting (GatewaySubnet and WorkloadSubnet)
 - VPN Gateway
 - Point-to-Site VPN Configuration
 - Linux Virtual Machine
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Part 2 — Video Presentation (20 Points)



Recording Rules

- **Camera ON**
- **Microphone ON**
- If you have issues → talk to me prior to the due date

Video Presentation

Your video must:

- **Be 5–7 minutes long**
- **Be well organized**
- **Include:**
 - **Introduction**
 - Introduce yourself
 - Confirm it is your own work
 - **Presentation**
 - Present your PowerPoint
 - Explain your architecture
 - Explain how Point-to-Site VPN provides secure access
 - Show screenshots of your Azure resources and settings
 - **Demo your project**

- Go through the checklist described in the demo section
 - **Reflection**
 - Share your personal experience from doing the project
 - What you have learned
 - What challenges you faced and how you solved them
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Part 3 — Demo Your Working Project (55 Points)

You must (Use as a checklist):

✓ Show VPN Connection from Local PC (20 Points)

- Open VPN client on your local computer
- Connect to Azure VPN
- Show status: **Connected**
- Show the Azure portal confirming an active P2S connection (if visible)

✓ Show SSH Access to VM via Private IP (20 Points)

- From your local PC (with VPN connected):
 - SSH to the Linux VM using its **private IP address**
 - Successfully log in
 - Run basic commands (hostname, whoami, ip addr)
 - Show that you are connected to the VM

✓ Show VPN Dependency (15 Points)

- **Disconnect the VPN** from your local PC
 - Attempt to SSH to the VM's private IP again
 - Show that SSH **fails** when VPN is disconnected
 - Explain why this proves the connection requires the VPN tunnel
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Penalties (Read Carefully ⚠)

You will lose marks if:

- Not showing **odl_user account** → **Grade = 0**
- Not showing **IP address spaces** → **50% penalty**
- Not using **name as prefix** for resources → **50% penalty**
- **Late submission** → **-10% per day**
- No **camera/voice** (without approval) → **Grade = 0**

How To Submit

- ✓ Record screen + camera showing your face + microphone
- ✓ Explain everything using your PowerPoint + demo while recording
- ✓ Keep video between **5–7 minutes**
- ✓ Upload the video as a **private file** to a streaming platform, like **YouTube** (publish as **Unlisted**)
- ✓ Submit your PowerPoint file and the video link in **Blackboard**

Clean Up Resources

Remember to **delete all created Azure resources** to avoid unwanted charges on your account credit.
